COURSE MATERIAL	
SUBJECT	BASIC ELECTRICAL & ELECTRONICS ENGINEERING(EE23AES101)
UNIT	3
COURSE	B.TECH
DEPARTMENT	ECE
SEMESTER	1-1
PREPARED BY (Faculty Name/s)	Dr M.Nagaiah Professor
Version	V-5
PREPARED / REVISED DATE	07-03-2021

TABLE OF CONTENTS – UNIT 3

1. Course Objectives

The objectives of the course are to make the students learn about:

1. To introduce basics of electric circuits.
2. To teach DC and AC electrical circuit analysis.
3. To explain working principles of transformers and electrical machines.
4. To impart knowledge on low voltage electrical installations

2. Prerequisites

Students should have knowledge on

1. Modern atomic theory.
2. Engineering Physics and Chemistry.

3. Syllabus

UNIT III

Basics of Power Systems:

Layout and operation of Hydro, Thermal, Nuclear Stations - Solar and Wind generating stations - Typical AC power supply scheme - Elements of Transmission line - Types of Distribution systems: Primary & Secondary distribution systems.

4. Course outcomes

1. Apply concepts of KVL/KCL in solving DC circuits (L3)
2. Choose correct rating of a transformer for a specific application (L5)
3. Illustrate working principles of induction motor - DC Motor (L3)
4. Identify type of electrical machine based on their operation. (L1)
5. Describe working principles of protection devices used in electrical circuits. (L2)

5. Co-PO / PSO Mapping

Machine Tools	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	3	2	2	3	2	2	3	2	3	2	3
CO2	3	2	3	3	2	3	3	2	3	3	2	2	3	3
CO3	3	3	3	3	3	3	3	3	3	3	3	2	3	3

CO4	3	3	3	3	3	3	3	3	3	3	3	3	3	3

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Layout & Operation of Hydro power Station	TB3/RB1
2		Layout & Operation of Thermal power Station	TB3/RB1
3		Principle & operation of Nuclear power Station	TB3/RB1
4		DC power generation through PV Cells	TB3/RB1
5		DC power generation with concentrated Solar plates	TB3/RB1
6	2	Types of wind Turbines (Horizontal & Vertical axis)	TB3/RB1
7		Operation of Wind Farm and generation of electric power	TB3/RB1
8		Typical AC power supply scheme and elements of Transmission System	TB3/RB1
9		Primary and Secondary distribution Systems	TB3/RB1

7. Activity Based Learning

1. Explaining circuits in laboratory for practical knowledge.
2. Mini project on designing basic circuits.

8. Lecture Notes

3.1 Hydroelectric Power Station:

A Generating station which utilizes the potential energy of water at a high level for the generation of electrical energy is known as a Hydroelectric Power Station.

Hydroelectric Power Station stations are generally located in hilly areas where dams can be built conveniently and large water reservoirs can be obtained. In a Hydroelectric Power Station, water head is created by constructing a dam across a river or lake. From the dam, water is led to a water turbine. The water turbine captures the energy in the falling water and changes the hydraulic energy (i.e., product of head and flow of water) into mechanical energy at the turbine shaft. The turbine drives the alternator which converts mechanical energy into electrical energy. Hydroelectric Power Station are becoming very popular because the reserves of fuels (i.e., coal and oil) are depleting day by day. They have the added importance for flood control, storage of water for irrigation and water for drinking purposes.

Advantages

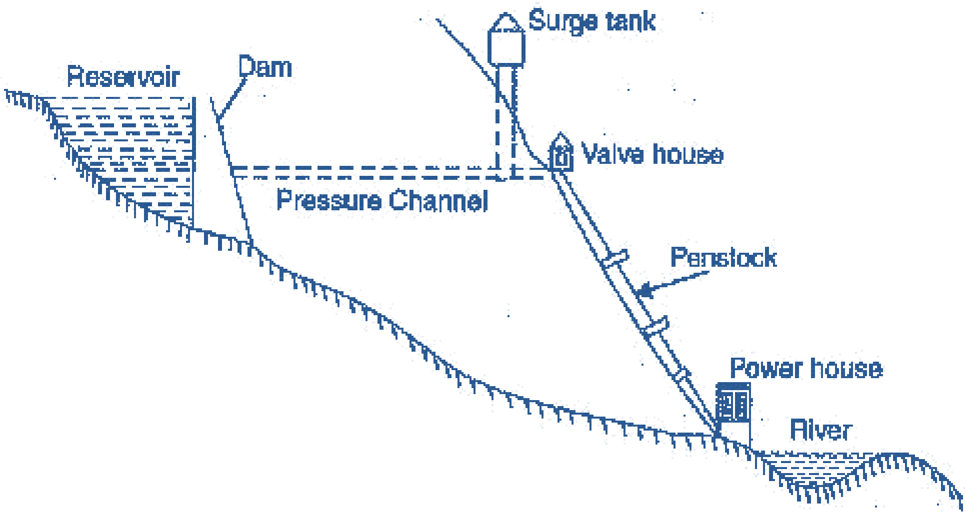
- It requires no fuel as water is used for the generation of electrical energy.
- It is quite neat and clean as no smoke or ash is produced.
- It requires very small running charges because water is the source of energy which is avail- able free of cost.
- It is comparatively simple in construction and requires less maintenance.
- It does not require a long starting time like a steam power station. In fact, such plants can be put into service instantly.
- It is robust and has a longer life.
- Such plants serve many purposes. In addition to the generation of electrical energy, they also help in irrigation and controlling floods.
- Although such plants require the attention of highly skilled persons at the time of construction, yet for operation, a few experienced persons may do the job well.

Disadvantages

- It involves high capital cost due to construction of dam.
- There is uncertainty about the availability of huge amount of water due to dependence on weather conditions.
- Skilled and experienced hands are required to build the plant.
- It requires high cost of transmission lines as the plant is located in hilly areas which are quite away from the consumers.

Schematic Arrangement of Hydroelectric Power Station

Although a Hydroelectric Power Station simply involves the conversion of hydraulic energy into electrical energy, yet it embraces many arrangements for proper working and efficiency. The schematic arrangement of. 2.2.



Schematic arrangement of a Hydro-electric plant

Fig. 2.2

The dam is constructed across a river or lake and water from the catchment area collects at the back of the dam to form a reservoir. A pressure tunnel is taken off from the reservoir and water brought to the valve house at the start of the penstock. The valve house contains main sluice valves and automatic isolating valves. The former controls the water flow to the power house and the latter cuts off supply of water when the penstock bursts, From the valve house, water is taken to water turbine through a huge steel pipe known as penstock. The water turbine converts hydraulic energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.

A surge tank (open from top) is built just before the valve house and protects the penstock from bursting in case the turbine gates suddenly close due to electrical load being thrown off. When the gates close, there is a sudden stopping of water at the lower end of the penstock and consequently the penstock can burst like a paper log. The surge tank absorbs this pressure swing by increase in its level of water.

Choice of Site for Hydro-electric Power Stations

The following points should be taken into account while selecting the site for a Hydroelectric Power Station:

- Availability of water. Since the primary requirement of a Hydroelectric Power Station is the availability of huge quantity of water, such plants should be built at a place (e.g., liver, canal) where adequate water is available at a good head.
- Storage of water. There are wide variations in water supply from a river or canal during the year. This makes it necessary to store water by constructing a dam in, order to ensure the generation of power throughout the year. The storage helps in equalizing the flow of water so that any excess quantity of water at a certain period of the year can be made available during times of very low flow in the river. This leads to the conclusion that site selected for a hydro-electric plant should provide adequate facilities for erecting a dam and storage of water.
- Cost and type of land. The land for the construction of the plant should be available at a reasonable price. Further, the bearing capacity of the ground should be adequate to with- stand the weight of heavy equipment to be installed.
- Transportation facilities. The site selected for a hydro-electric plant should be accessible by rail and road so that necessary equipment and machinery could be easily transported.

It is clear from the above-mentioned factors that ideal choice of site for such a plant is near a river in hilly areas where dam can be conveniently built and large reservoirs can be obtained.

Constituents of Hydro-electric Plant

The constituents of a hydro-electric plant are (1) hydraulic structures (2) water turbines and
(3) electrical equipment. We shall discuss these items in turn.

1. Hydraulic structures. Hydraulic structures in a Hydroelectric Power Station include darn, spillways, headworks, surge tank, penstock and accessory works.

(1) Dam. A dam is a barrier which stores water and creates water head. Dams are built of concrete or stone masonry, earth or rock fill. The type and arrangement depend upon
the topography of the site. A masonry dam may be built in a narrow canyon. An earth dam may be best suited for a wide valley. The type of dam also depends upon the foundation conditions, local materials and transportation available, occurrence of earthquakes and other hazards. At most of sites, more than one type of dam may be suitable and the one which is most economical is chosen.

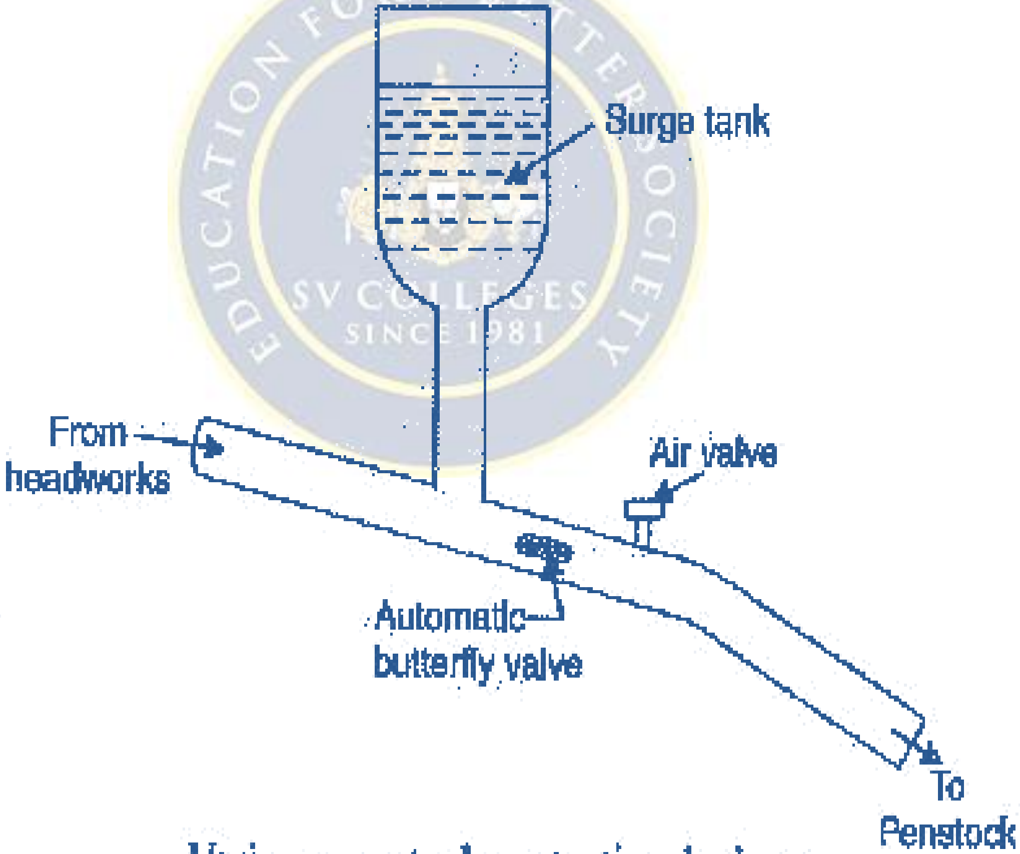
(2) Spillways. There are times when the river flow exceeds the storage capacity of the reservoir. Such a situation arises during heavy rainfall in the catchment area. In order to discharge the surplus water from the storage reservoir into the river on the down-stream side of the dam, spillways are used. Spillways are constructed of concrete piers on the top of the dam. Gates are provided between these piers and surplus water is discharged over the crest of the dam by opening these gates.

(3) Headworks. The headworks consists of the diversion structures at the head of an intake. They generally include booms and racks for diverting floating debris, sluices for by-passing debris and sediments and valves for controlling the flow of water to the turbine. The flow of water into and though headworks should be as smooth as possible to avoid head loss and cavitation. For this purpose, it is necessary to avoid sharp corners and abrupt contractions or enlargements.

(4) Surge tank. Open conduits leading water to the turbine require no protection. However, when closed conduits are used, protection becomes necessary to limit the abnormal pressure in the dam. For this reason, closed conduits are always provided with a surge tank. A surge tank is a small reservoir or tank (open at the top) in which water level rises or falls to reduce the pressure swings in the conduit.

A surge tank is located near the beginning of the conduit. When the turbine is running at a steady load, there are no surges in the flow of water through the conduit i.e., the quantity of water flowing in the conduit is just sufficient to meet the turbine requirements. However, when the load on the turbine decreases, the governor closes the gates of turbine, reducing water supply to the turbine. The excess water at the lower end of the conduit rushes back to the surge tank and increases its water level. Thus, the conduit is prevented from bursting. On the other hand, when load on the turbine increases, additional water is drawn from the surge tank to meet the increased load requirement. Hence, a surge tank overcomes the abnormal pressure in the conduit when load on the turbine falls and acts as a reservoir during increase of load on the turbine.

(5) Penstocks. Penstocks are open or closed conduits which carry water to the turbines. They are generally made of reinforced concrete or steel. Concrete penstocks are suitable for low heads (< 30 m) as greater pressure causes rapid deterioration of concrete. The steel pen- stocks can be designed for any head; the thickness of the penstock increases with the head or working pressure.



Various penstock protective devices

Fig. 2.3

Various devices such as automatic butterfly valve, air valve and surge tank (See Fig. 2.3) are provided for the protection of penstocks. Automatic butterfly valve shuts off water flow through the penstock promptly if it ruptures. Air valve maintains the air pressure inside the penstock equal to outside atmospheric pressure. When water runs out of a penstock faster than it enters, a vacuum is created which may cause the penstock to collapse. Under such situations, air valve opens and admits air in the penstock to maintain inside air pressure equal to the outside air pressure.

Water turbines. Water turbines are used to convert the energy of falling water into mechanical energy. The principal types of water turbines are:

(i) Impulse turbines (ii) Reaction turbines

(1) Impulse turbines. Such turbines are used for high heads. In an impulse turbine, the entire pressure of water is converted into kinetic energy in a nozzle and the velocity of the jet drives the wheel. The example of this type of turbine is the Pelton wheel (See Fig. 2.4). It consists of a wheel fitted with elliptical Operating buckets along its periphery. The force of water jet striking the buckets on the wheel drives the turbine. The quantity of water jet falling on the turbine is controlled by means of a needle or spear (not shown in the figure) placed in the tip of the nozzle. The movement of the needle is controlled by the governor. If the load on the turbine decreases, the governor pushes the needle into the nozzle, thereby reducing the quantity of water striking the buckets. Reverse action takes place if the load on the turbine increases.

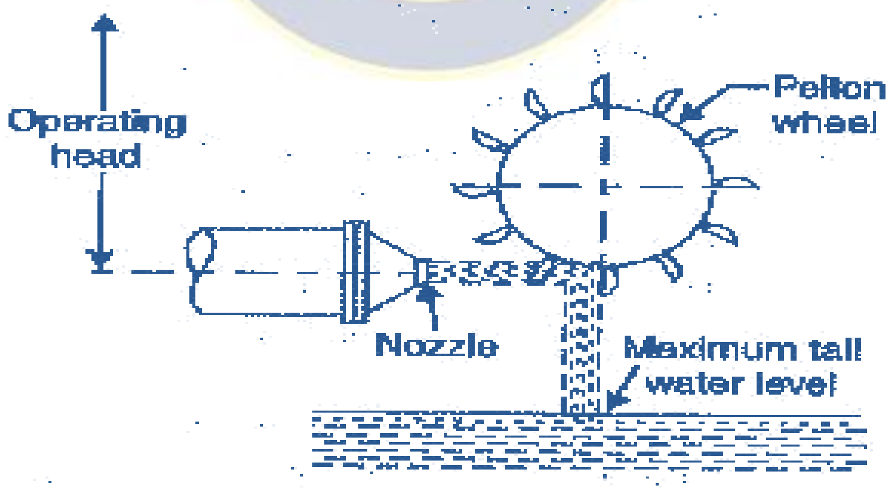


Fig. 2.4 Pelton Wheel

(ii) Reaction turbines. Reaction turbines are used for low and medium heads. In a reaction turbine, water enters the runner partly with pressure energy and partly with velocity head. The important types of reaction turbines are :

(a) Francis turbines (b) Kaplan turbines

A Francis turbine is used for low to medium heads. It consists of an outer ring of stationary guide blades fixed to the turbine casing and an inner ring of rotating blades forming the runner. The guide blades control the flow of water to the turbine. Water flows radially inwards and changes to a downward direction while passing through the runner. As the water passes over the "rotating blades" of the runner, both pressure and velocity of water are reduced. This causes a reaction force which drives the turbine.

A Kaplan turbine is used for low heads and large quantities of water. It is similar to Francis turbine except that the runner of Kaplan turbine receives water axially. Water flows radially inwards through regulating gates all around the sides, changing direction in the runner to axial flow. This causes a reaction force which drives the turbine.

(iii). Electrical equipment. The electrical equipment of a hydro-electric power station includes alternators, transformers, circuit breakers and other switching and protective devices.

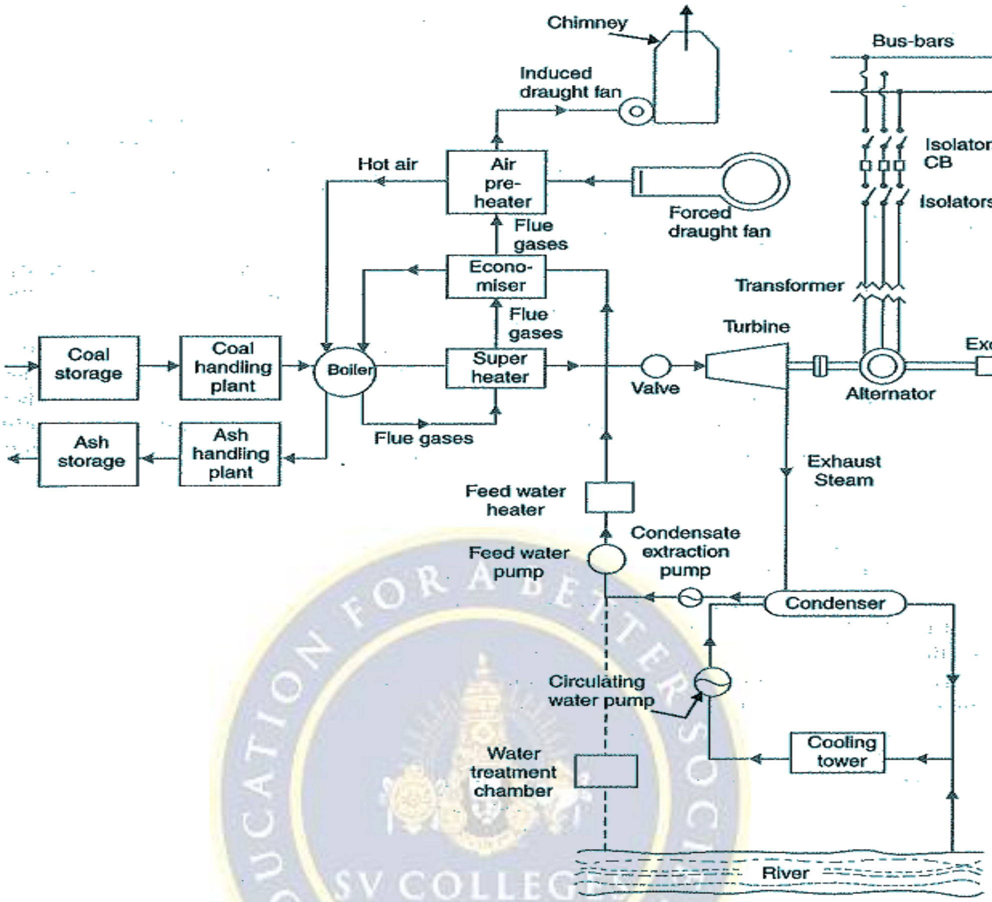
3.2 THERMAL POWER STATION

The heat released during the combustion of coal, oil or gas is used in a boiler to raise steam. In India heat generation is mostly coal based except in small sites, because of limited indigenous production of oil. Therefore, we shall discuss only coal fired boilers for steam turbine in order to produce electrical energy.

The chemical energy stored in coal is transformed into electric energy in thermal power plants. The heat released by the combustion of coal produces steam in a boiler at high pressure and temperature, which when passed through a steam turbine gives off some of its internal energy as mechanical energy. The axial flow type of turbine is normally used with several cylinders on the same shaft. The steam turbine acts as a prime mover and drives the electric generator (alternator).

Schematic Arrangement of Steam Power Station:

Although steam power station simply involves the conversion of heat of coal combustion into electrical energy, yet it embraces many arrangements for proper working and efficiency. The schematic arrangement of a modern steam power station is shown in Fig. 2.1. The whole arrangement can be divided into the following stages for the sake



Schematic arrangement of Steam Power Station

Fig. 2.1

of simplicity:

1. Coal and ash handling arrangement
2. Steam generating plant
3. Steam turbine
4. Alternator
5. Feed water
6. Cooling arrangement

1. Coal and ash handling plant:The coal is transported to the power station by road or rail and is stored in the coal storage plant. Storage of coal is primarily a matter of protection against coal strikes, failure of transportation system and general coal shortages. From the coal storage plant, coal is delivered to the coal handling plant where it is pulverized (i.e., crushed

into small pieces) in order to increase its surface exposure, thus promoting rapid combustion without using large quantity of excess air. The pulverized coal is fed to the boiler by belt conveyors. The coal is burnt in the boiler and the ash produced after the complete combustion of coal is removed to the ash handling plant and then delivered to the ash storage plant for disposal. The removal of the ash from the boiler furnace is necessary for proper burning of coal.

It is worthwhile to give a passing reference to the amount of coal burnt and ash produced in a modern thermal power station. A 100 MW station operating at 50% load factor may burn about 20,000 tons of coal per month and ash produced may be to the tune of 10% to 15% of coal fired i.e., 2,000 to 3,000 tons. In fact, in a thermal station, about 50% to 60% of the total operating cost consists of fuel purchasing and its handling.

2.Steam generating plant: The steam generating plant consists of a boiler for the production of steam and other auxiliary equipment for the utilisation of flue gases.

Boiler:The heat of combustion of coal in the boiler is utilised to convert water into steam at high temperature and pressure. The flue gases from the boiler make their journey through superheater, economiser, air pre-heater and are finally exhausted to atmosphere through the chimney.

Superheater: The steam produced in the boiler is wet and is passed through a superheater where it is dried and superheated (i.e., steam temperature increased above that of boiling point of water) by the flue gases on their way to chimney. Superheating provides two principal benefits. Firstly, the overall efficiency is increased. Secondly, too much condensation in the last stages of turbine (which would cause blade corrosion) is avoided. The superheated steam from the superheater is fed to steam turbine through the main valve.

Economiser: An economiser is essentially a feed water heater and derives heat from the flue gases for this purpose. The feed water is fed to the economiser before supplying to the boiler. The economiser extracts a part of heat of flue gases to increase the feed water temperature.

Air preheater: An air preheater increases the temperature of the air supplied for coal burning by deriving heat from flue gases. Air is drawn from the atmosphere by a forced draught fan and is passed through air preheater before supplying to the boiler furnace. The air preheater extracts heat from flue gases and increases the temperature of air used for coal combustion.

The principal benefits of preheating the air are: increased thermal efficiency and increased steam capacity per square metre of boiler surface.

3. Steam turbine: The dry and super-heated steam from the super heater is fed to the steam turbine through main valve. The heat energy of steam when passing over the blades of turbine is converted into mechanical energy. After giving heat energy to the turbine, the steam is exhausted to the condenser which condenses the exhausted steam by means of cold water circulation.
4. Alternator: The steam turbine is coupled to an alternator. The alternator converts mechanical energy of turbine into electrical energy. The electrical output from the alternator is delivered to the bus bars through transformer, circuit breakers and isolators.
5. Feed water. The condensate from the condenser is used as feed water to the boiler. Some water may be lost in the cycle which is suitably made up from external source. The feed water on its way to the boiler is heated by water heaters and economizer. This helps in raising the overall efficiency of the plant.
6. Cooling arrangement: In order to improve the efficiency of the plant, the steam exhausted from the turbine is condensed by means of a condenser. Water is drawn from a natural source of supply such as a river, canal or lake and is circulated through the condenser. The circulating water takes up the heat of the exhausted steam and itself becomes hot. This hot water coming out from the condenser is discharged at a suitable location down the river. In case the availability of water from the source of supply is not assured throughout the year, cooling towers are used. During the scarcity of water in the river, hot water from the condenser is passed on to the cooling towers where it is cooled. The cold water from the cooling tower is reused in the condenser.

Choice of Site for Steam Power Stations

In order to achieve overall economy, the following points should be considered while selecting a site for a steam power station:

1. Supply of fuel: The steam power station should be located near the coal mines so that transportation cost of fuel is minimum. However, if such a plant is to be installed at a place where coal is not available, then care should be taken that adequate facilities exist for the transportation of coal.
2. Availability of water: As huge amount of water is required for the condenser, therefore, such a plant should be located at the bank of a river or near a canal to ensure the continuous supply of water.Transportation facilities: A modern steam power station often requires the transportation

of material and machinery. Therefore, adequate transportation facilities must exist e., the plant should be well connected to other parts of the country by rail, road. etc.

3. Cost and type of land: The steam power station should be located at a place where land is cheap and further extension, if necessary, is possible. Moreover, the bearing capacity of the ground should be adequate so that heavy equipment could be installed.
4. Nearness to land centres: In order to reduce the transmission cost, the plant should be located near the centre of the load. This is particularly important if d.c. supply system is adopted. However, if a.c. supply system is adopted, this factor becomes relatively less important: It is because a.c. power can be transmitted at high voltages with consequent reduced transmission cost. Therefore, it is possible to install the plant away from the load centres, provided other conditions are favourable.
5. Distance from populated area: As huge amount of coal is burnt in a steam power station, therefore, smoke and fumes pollute the surrounding area. This necessitates that the plant should be located at a considerable distance from the populated areas.

Conclusion: It is clear that all the above factors cannot be favourable at one place. However, keeping in view the fact that now-a-days the supply system is a.c. and more importance is being given to generation than transmission, a site away from the towns may be selected. In particular, a site by river side where sufficient water is available, no pollution of atmosphere occurs and fuel can be transported economically, may perhaps be an ideal choice.

3.3 Nuclear Power Station:

A generating station in which nuclear energy is converted into electrical energy is known as a nuclear power station.

In nuclear power station, heavy elements such as Uranium (U235) or Thorium (Th232) are sub- jected to nuclear fission in a special apparatus known as a reactor. The heat energy thus released is utilised in raising steam at high temperature and pressure. The steam runs

the steam turbine which converts steam energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.

The most important feature of a nuclear power station is that huge amount of electrical energy can he produced from a relatively small amount of nuclear fuel as compared to other conventional types of power stations. It has been found that complete fission of 1 kg of Uranium (U235) can produce as much energy as can be produced by the burning of 4,500 tons of high-grade coal. Although the recovery of principal nuclear fuels (i.e., Uranium and Thorium) is difficult and expensive, yet the total energy content of the estimated world reserves of these fuels are considerably higher than those of conventional fuels, viz., coal, oil and gas. At present, energy crisis is gripping-us and, therefore, nuclear energy can be successfully employed for producing low cost electrical energy on a large scale to meet the growing commercial and industrial demands.

Advantages

- The amount of fuel required is quite small. Therefore, there is a considerable saving in the cost of fuel transportation.
- A nuclear power plant requires less space as compared to any other type of the same size.
- It has low running charges as a small amount of fuel is used for producing bulk electrical energy.
- This type of plant is very economical for producing bulk electric power.
- It can be located near the load centres because it does not require large quantities of water and need not be near coal mines. Therefore, the cost of primary distribution is reduced.
- There are large deposits of nuclear fuels available all over the world. Therefore, such plants can ensure continued supply of electrical energy for thousands of years.
- It ensures reliability of operation.

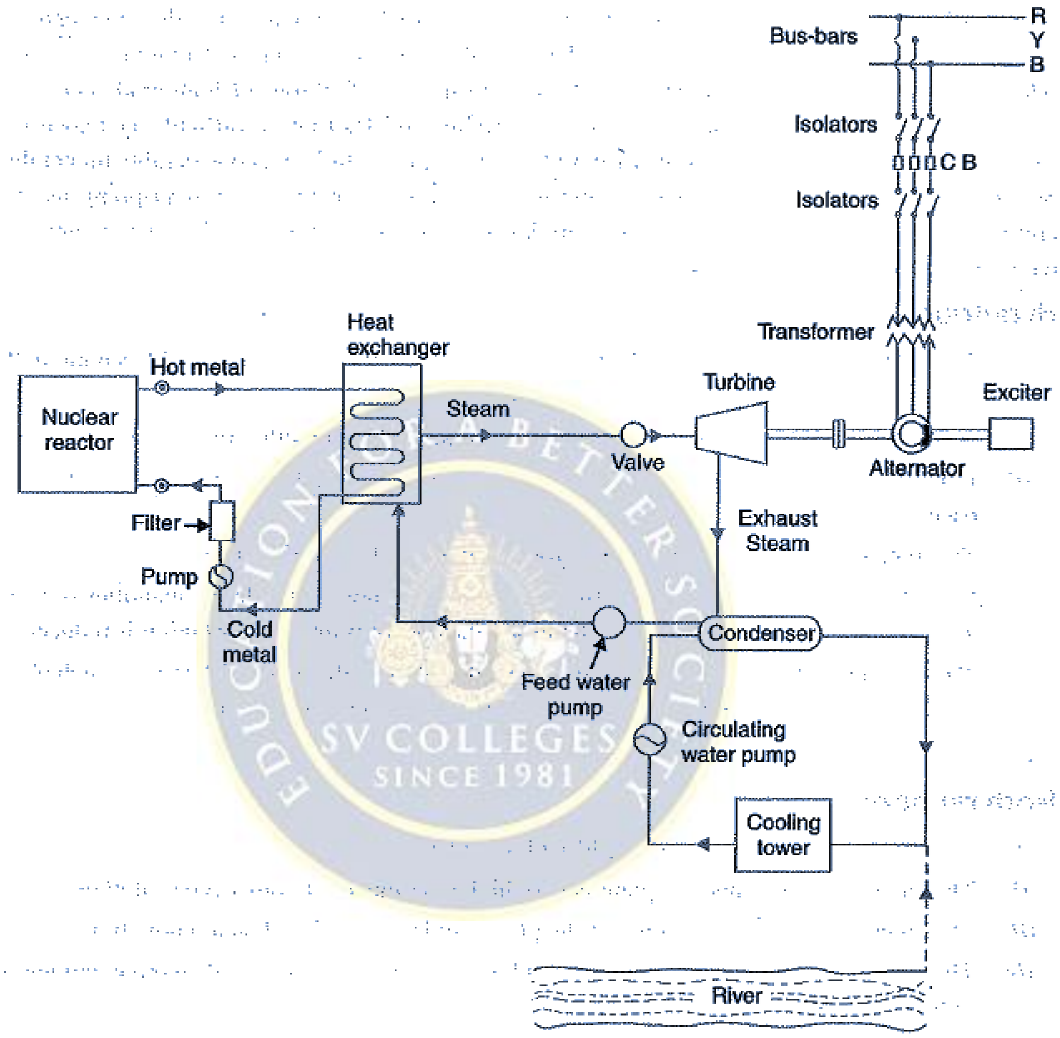
Disadvantages

- The fuel used is expensive and is difficult to recover.
- The capital cost on a nuclear plant is very high as compared to other types of plants.
- The erection and commissioning of the plant requires greater technical knowledge.
- The fission by-products are generally radioactive and may cause a dangerous amount of radioactive pollution.
- Maintenance charges are high due to lack of standardisation. Moreover, high salaries of specially trained personnel employed to handle the plant further raise the cost.
- Nuclear power plants are not well suited for varying loads as the reactor does not respond to the load fluctuations efficiently.
- The disposal of the by-products, which are radioactive, is a big problem. They have either to be disposed off in a deep trench or in a sea away from sea-shore.

Schematic Arrangement of Nuclear Power Station

The schematic arrangement of a nuclear power station is shown in Fig. 2.7. The whole arrangement can be divided into the following main stages :

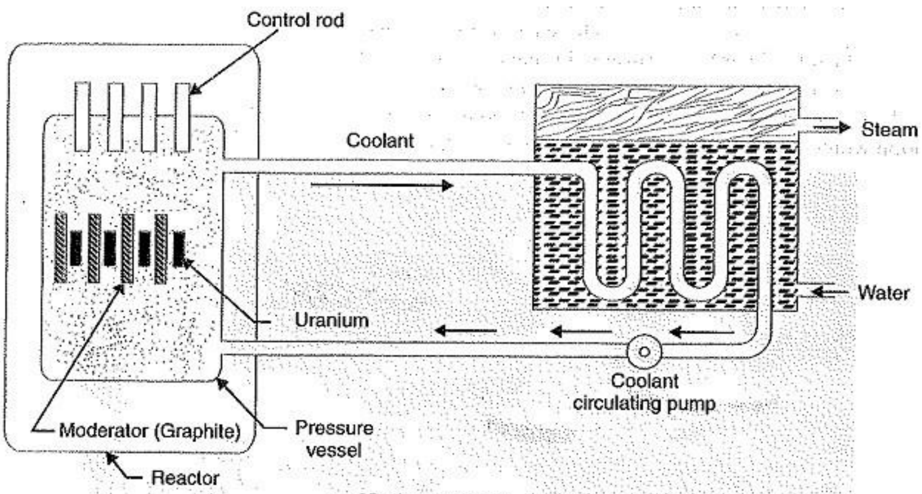
(i) Nuclear reactor (ii) Heat exchanger (iii) Steam turbine (iv) Alternator



Schematic arrangement of Nuclear Power Station

Fig. 2.7

(1) Nuclear reactor. It is an apparatus in which nuclear fuel (U235) is subjected to nuclear fission. It controls the chain reaction* that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released.



Nuclear Reactor
Fig. 2.8

(2) Heatexchanger. The coolant gives up heat to the heat exchanger which is utilised in raising the steam. After giving up heat, the coolant is again fed to the reactor.

(3) Steam Turbine.The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted

to condenser. The condenser condenses the steam which is fed to the heat exchanger through feed water pump.

(4) Alternator. The steam turbine drives the alternator which converts mechanical energy into electrical energy. The output from the alternator is delivered to the bus-bars through transformer, circuit breakers and isolators.

Selection of Site for Nuclear Power Station

The following points should be kept in view while selecting the site for a nuclear power station

A nuclear reactor is a cylindrical stout pressure vessel and houses fuel rods of Uranium, moderator and control rods (See Fig. 2.8). The fuel rods constitute the fission material and release huge amount of energy when bombarded with slow moving neutrons. The moderator consists of graphite rods which enclose the fuel rods. The moderator slows down the neutrons before they bombard the fuel rods. The control rods are of cadmium and are inserted into the reactor. Cadmium is strong neutron absorber and thus regulates the supply of neutrons for fission. When the control rods are pushed in deep enough they absorb most of fission neutrons and hence few are available for chain reaction which, therefore, stops. However, as they are being withdrawn, more and more of these fission neutrons cause fission and hence the intensity of chain reaction (or heat produced) is increased. Therefore, by pulling out the control rods, power of the nuclear reactor is increased, whereas by pushing them in, it is reduced. In actual practice, the lowering or raising of control rods is accomplished automatically according to the requirement of load. The heat produced in the reactor is removed by the coolant, generally a sodium metal. The coolant carries the heat to the heat exchanger.

- Availability of water. As sufficient water is required for cooling purposes, therefore, the plant site should be located where ample quantity of water is available, g., across a river or by sea-side.
- Disposal of waste. The waste produced by fission in a nuclear power station is generally radioactive which must be disposed off properly to avoid health hazards. The waste should either be buried in a deep trench or disposed off in sea quite away from the sea shore. Therefore, the site selected for such a plant should have adequate arrangement for the dis-posal of radioactive waste.
- Distance from populated areas. The site selected for a nuclear power station should be quite away from the populated areas as there is a danger of presence of radioactivity in the atmosphere near the plant. However, as a precautionary measure, a dome is used in the plant which does not allow the radioactivity to spread by wind or underground waterways.
- Transportation facilities. The site selected for a nuclear power station should have adequate facilities in order to transport the heavy equipment during erection and to facilitate the movement of the workers employed in the plant.

3.4 WIND GENERATING STATION

We are well aware how important it is to make use of renewable energy resources as much as we can. Conventional energy resources, such as fossil fuels, are exhaustible and may get vanished someday if we keep using them at the current rate. That is why technologies are being developed to harvest electrical energy efficiently and reliably from renewable energy sources such as hydropower, solar power, wind power, geothermal power etc. Wind is one of the important renewable resources we could make use of. Human civilizations have been using wind energy from thousands of years. Earlier windmills were used to pump up the water from a well or to crush grains, but the scenario has changed a lot and we are using windmills to generate electricity. This article explains how electricity is generated from wind energy i.e. wind power.

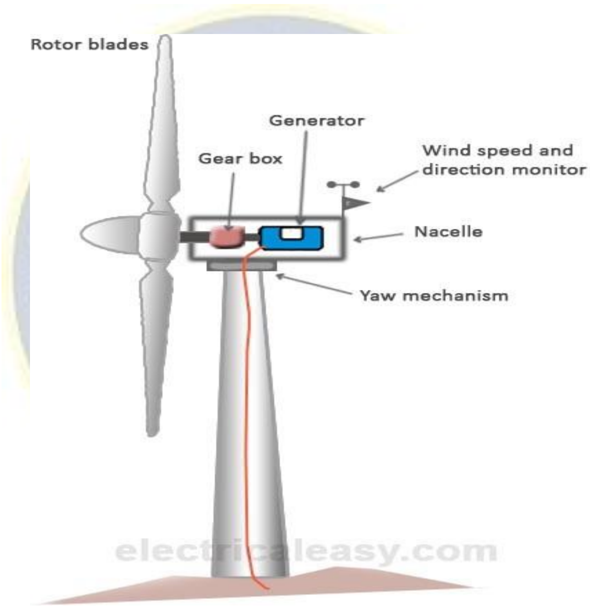
How Wind Power Works?

Wind possesses kinetic energy which can be converted into electrical energy. For converting the kinetic energy of wind into electricity, wind turbines are employed. The wind rotates the turbine blades, the shaft of which is mechanically coupled to an electric generator. The basic working principle of a wind turbine is as simple as it sounds! But, yes, for harnessing maximum energy from the wind, wind turbines in real are a bit sophisticated. They are categorized between two main types as - Horizontal Axis Wind Turbine (HAWT) and Verticle Axis Wind Turbine (VAWT). They are also available in various sizes and ratings.
You can also find a rooftop wind turbine which can sit on the roof of your home and contributing to your electricity needs. On the other hand, a large horizontal axis wind turbine can be taller than 300 feet. It's not only about a single home, but there are wind power plants (aka wind farms) which are connected to electrical power grid. Many countries are trying to increase the share of wind power into their total generation.

A wind farm (or wind power plant) consists of many individual wind turbines grouped together. There are many examples of large wind farm installations of about a couple of thousand MW capacity over the world. A wind farm can also be located off-shore. The location is selected on the basis of availability and average speed of the wind at that location. Thus, many times these are located at higher altitudes.

3.4.1 HAWT (Horizontal Axis Wind Turbine)

Horizontal axis wind turbines are most commonly employed in a wind farm. The figure below shows various basic components of a horizontal axis wind turbine.



- Tower: The tower is usually cylindrical with the heights ranging from 25 meters up to 90 meters. An arrangement is made inside the tower to climb up at the top for maintenance purposes.

- Rotor blades: Usually three-bladed wind turbines are used in wind farms. The length of rotor blades ranges from 20 to 40 meters. They are made from a light-weight material such as fiberglass reinforced polyester or wood epoxy. When wind is blown, the rotor blades rotate at a speed between 10 to 60 RPM.
- Nacelle: It is mounted at the top of the tower and it houses the gearbox and generator assembly. It also has wind speed and direction monitoring mechanism attached.
- Gearbox: It takes lower RPM from the rotor blades and provides higher RPM to the generator through the gear's arrangement. (There are direct drive wind turbines too in which gearbox is absent.)
- Generator: Rotor assembly is coupled to the generator through the gearbox. Thus, when the rotor rotates it drives the generator and, hence, electricity is generated.
- Yaw mechanism: For an increased efficiency, rotor blades must face the wind direction. But, as the wind direction changes time to time it makes sense to turn the rotor assembly to face the changed wind direction. Yaw mechanism does that!

3.5 SOLAR GENERATING STATION

Needless to say that the Sun is the biggest source of renewable energy for the Earth. The fact is that even though the earth receives only a part of the energy generated by the Sun (i.e.
Solar energy), that part of solar energy is also tremendously huge. The Earth receives solar energy in the form of light and heat. But in today's world, the words 'power' and 'energy' are leaned more towards 'electricity'. This article explains how electricity is harvested from the solar energy and how it is utilized.

How solar power works?

Electrical energy can be harvested from solar power by means of either photovoltaics or concentrated solar power systems.

Photovoltaics (PV)

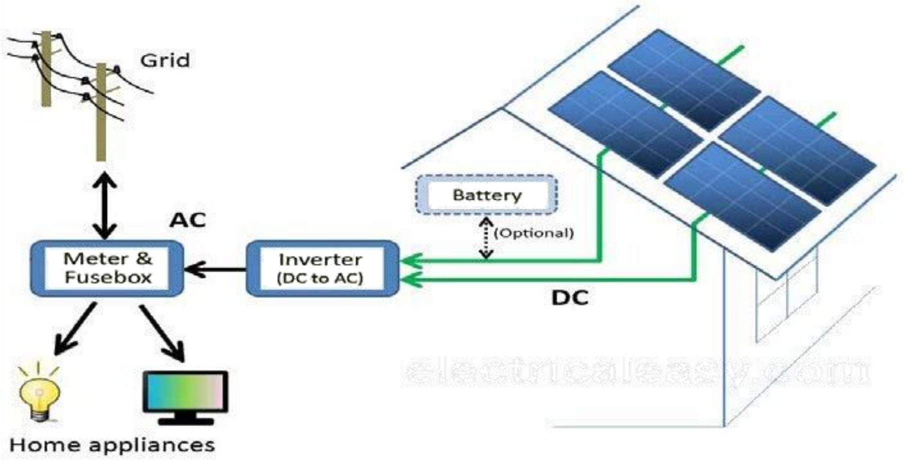
Photovoltaics directly convert solar energy into electricity. They work on the principle of the photovoltaic effect. When certain materials are exposed to light, they absorb photons and release free electrons. This phenomenon is called as the photoelectric effect. Photovoltaic effect is a method of producing direct current electricity based on the principle of the photoelectric effect.

Based on the principle of photovoltaic effect, solar cells or photovoltaic cells are made. They convert sunlight into direct current (DC) electricity. But, a single photovoltaic cell does not produce enough amount of electricity. Therefore, a number of photovoltaic cells are mounted on a supporting frame and are electrically connected to each other to form a photovoltaic module or solar panel. Commonly available solar panels range from several hundred watts (say 100 watts) up to few kilowatts (ever heard of a 5kW solar panel?). They are available in different sizes and different price ranges. Solar panels or modules are designed to supply electric power at a certain voltage (say 12v), but the current they produce is directly dependent on the incident light. As of now it is clear that photovoltaic modules produce DC electricity. But, for most of the times we require AC power and, hence, solar power system consists of an inverter too.

3.5.1 Photovoltaic solar power system

According to the requirement of power, multiple photovoltaic modules are electrically connected together to form a PV array and to achieve more power. There are different types of PV systems according to their implementation.

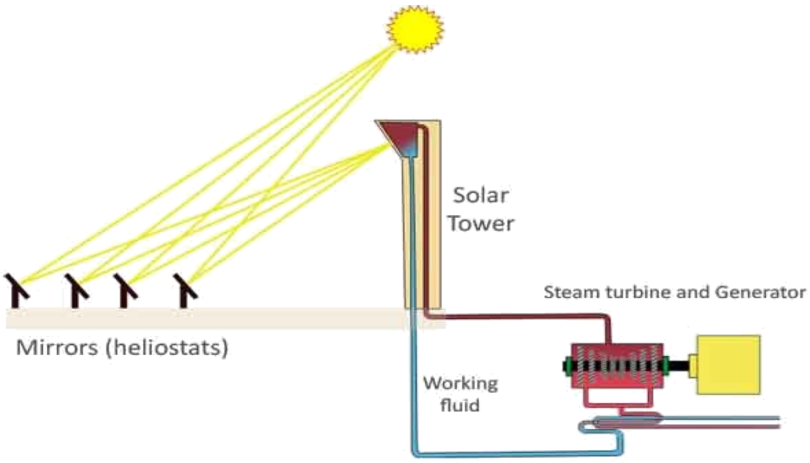
- PV direct systems: These systems supply the load only when the Sun is shining. There is no storage of power generated and, hence, batteries are absent. An inverter may or may not be used depending on the type of load.
- Off-grid systems: This type of system is commonly used at locations where power from the grid is not available or not reliable. An off-grid solar power system is not connected to any electric grid. It consists solar panel arrays, storage batteries and inverter circuits.
- Grid connected systems: These solar power systems are tied with grids so that the excess required power can be accessed from the grid. They may or may not be backed by batteries.



Typical grid connected PV solar system

3.5.2 Concentrated solar power

As the name suggest, in this type of solar power system, sun rays are concentrated (focused) on a small area by placing mirrors or lenses over a large area. Due to this, a huge amount of heat is generated at the focused area. This heat can be used to heat up the working fluid which can further drive the steam turbine. There are different types of technologies that are based on the concentrated solar power to produce electricity. Some of them are - parabolic trough, Stirling dish, solar power tower etc. The following schematic shows how a solar power tower works.



Solar Power Tower

3.6 Electric Supply System

The conveyance of electric power from a power station to consumers' premises is known as Electric Supply System.

An electric supply system consists of three principal components viz., the power station, the transmission lines and the distribution system. Electric power is produced at the power stations which are located at favorable places, generally quite away from the consumers. It is then transmitted over large distances to load centres with the help of conductors known as transmission lines. Finally, it is distributed to a large number of small and big consumers through a distribution network.

The electric supply system can be broadly classified into

(i) d.c. or a c. system

(ii) overhead or underground system.

Nowadays, 3-phase, 3-wire a.c. system is universally adopted for generation and transmission of electric power as an economical proposition. However, distribution of electric power is done by 3-phase, 4-wire a.c. system. The underground system is more expensive than the overhead system. Therefore, in our country, overhead system is mostly adopted for transmission and distribution of electric power.

Typical A.C. Power Supply Scheme:

The large network of conductors between the power station and the consumers can be broadly divided into two parts viz., transmission system and distribution system. Each part can be further sub-divided into two primary transmission and secondary transmission and primary distribution and secondary distribution. Fig. 7.1. shows the layout of a typical a.c. power supply scheme by a single line diagram. It may be noted that it is not necessary that all power schemes include all the stages shown in the figure. For example, in a certain power scheme, there may be no secondary transmission and, in another case, the scheme may be so small that there is only distribution and no transmission.

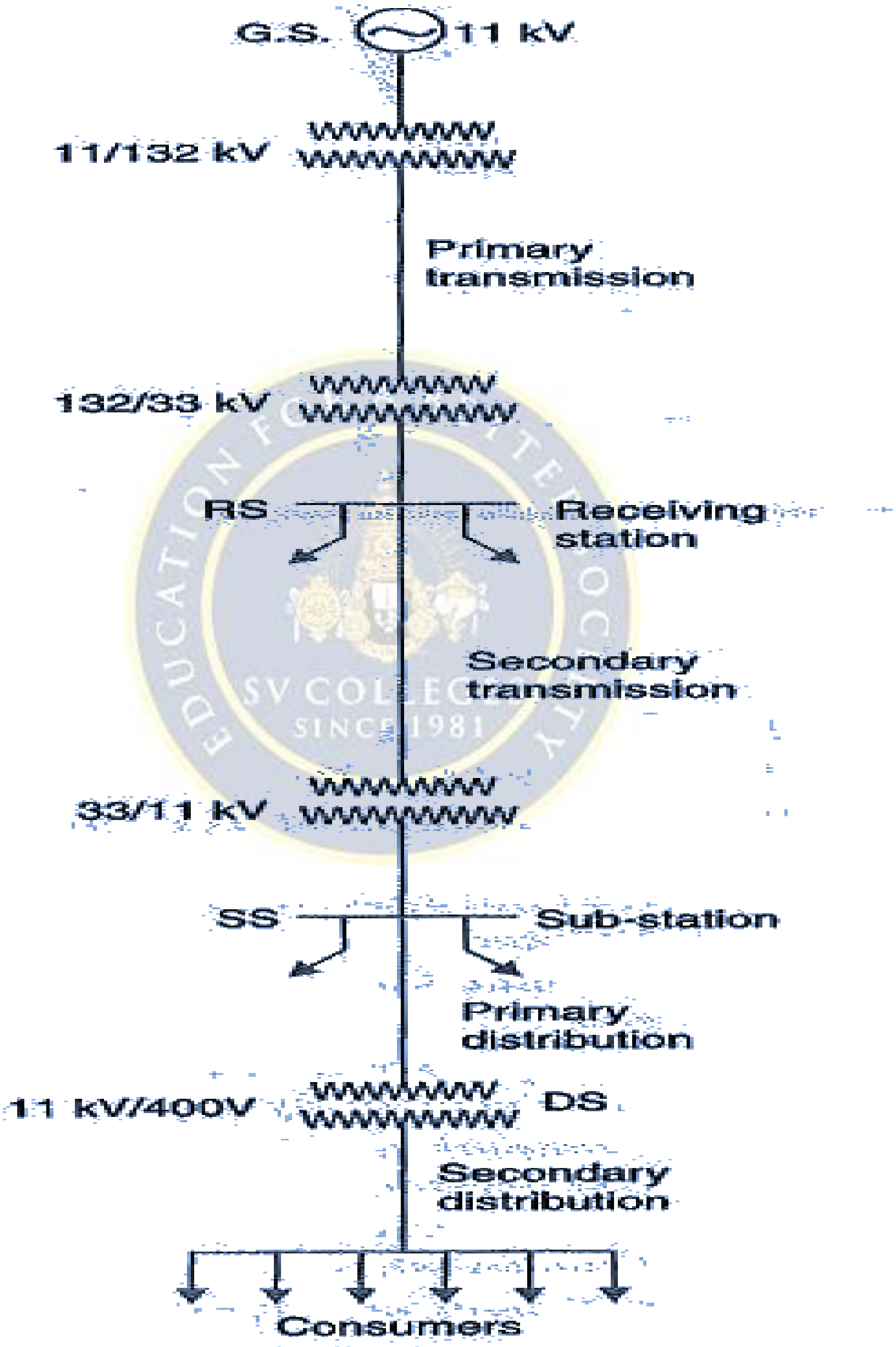


Fig. 7.1

1. Generating station: In Fig 7.1, G.S. represents the generating station where electric power is produced by 3-phase alternators operating in parallel. The usual generation voltage is 11 kV. For economy in the transmission of electric power, the generation voltage (i.e., 11 kV) is stepped upto 132 kV (or more) at the generating station with the help of 3-phase trans- formers. The transmission of electric power at high voltages has several advantages including the saving of conductor material and high transmission efficiency. It may appear advisable to use the highest possible voltage for transmission of electric power to save conductor material and have other advantages. But there is a limit to which this voltage can be increased. It is because increase in transmission voltage introduces insulation problems as well as the cost of switchgear and transformer equipment is increased. Therefore, the choice of proper transmission voltage is essentially a question of economics. Generally the primary transmission is carried at 66 kV, 132 kV, 220 kV or 400 kV.
2. Primary transmission: The electric power at 132 kV is transmitted by 3-phase, 3-wire overhead system to the outskirts of the city. This forms the primary transmission.
3. Secondary transmission: The primary transmission line terminates at the receiving station (RS) which usually lies at the outskirts of the city. At the receiving station, the voltage is re duced to 33kV by step-down transformers. From this station, electric power is transmitted at 33kV by 3-phase, 3-wire over¬head system to various sub-stations (SS) located at the strategic points in the city. This forms the secondary transmission.
4. Note. A practical power system has large number of auxiliary equipments (e.g., fuses, circuit breakers, voltage control devices etc.). However, such equipments are not shown in Fig. 7.1. It is because the amount of information included in the diagram depends on the purpose for which the diagram is intended. Here our purpose is to display general lay out of the powersystem. Therefore, the location of circuit breakers, relays etc., is unimportant.

   Further, the structure of power system is shown by a single line diagram. The complete 3- phase circuit is seldom necessary to convey even the most detailed information about the system. In fact, the complete diagram is more likely to hide than to clarify the information we are seeking from the system viewpoint.

3.7 Main components of a transmission line

Due to the economic considerations, three-phase three-wire overhead system is widely used for electric power transmission. Following are the main elements of a typical power system.

- Conductors: three for a single circuit line and six for a double circuit line. Conductors must be of proper size (i.e. cross-sectional area). This depends upon its current capacity. Usually, ACSR (Aluminium-core Steel-reinforced) conductors are used.

- Transformers: Step-up transformers are used for stepping up the voltage level and step- down transformers are used for stepping it down. Transformers permit power to be transmitted at higher efficiency.
- Line insulators: to mechanically support the line conductors while electrically isolating them from the support towers.
- Support towers: to support the line conductors suspending in the air overhead.
- Protective devices: to protect the transmission system and to ensure reliable operation. These include ground wires, lightening arrestors, circuit breakers, relays etc.
- Voltage regulators: to keep the voltage within permissible limits at the receiving end.

9. Practice Quiz

1. Which of the following is not an advantage of hydroelectric power plant?[b]
   a) no fuel requirement b) low running cost
   c) continuous power source d) no standby losses

2. What is the principle of operation of steam power plant? [d]
   a) Carnot cycle b) Brayton cycle
   c) Stirling cycle d) Rankine cycle

3. Which of the following is a part of air and fuel gas circuit? [b]
   a) Condenser b) Economiser
   c) Air preheater c) Cooling tower

4. The best capable alternative source which can meet the future energy demand [b]
   a) thermal power plant b) nuclear power plant
   c) hydroelectric power plant d) geothermal power plant

5. How much coal is required to generate energy equivalent to the energy generated by 1 kg of uranium? [d]
   a) 30000 tonnes of high grade coal
   b) 300 tonnes of high grade coal
   c) 10000 tonnes of high grade coal
   d) 3000 tonnes of high grade coal

6. What is the overall efficiency of nuclear power plant? [c]
   a) 20 to 25%
   b) 25 to 30%
   c) 30 to 40 %
   d) 50 to 70 %

7. The function of super heater to increase…. [b]

a) pressure b) temperature
c)both d) none

8.In thermal power plant, more heat loss occur at [a]

a) Condenser b) Economiser
c) Air preheater d) Cooling tower

9.which one is converts the mechanical energy into electrical energy [d]

a) tutbine b) motor
c) Air preheater d) alternator

10.the turbine output is……. energy [c]

a) electrical b) steam
c) mechanical d) water

11. Nuclear reactor generally employs [a]

a) fission b) fusion
c) both d) none

12. the purpose of control rods in nuclear reaction is [b]

a) slow down fast neutrons b) absorbing neutrons
c) to reflect neutrons d) none

13. the purpose of moderators in nuclear reaction is [a]

a) slow down fast neutrons b) absorbing neutrons
c) to reflect neutrons d) none

14. the material used for control rod is [c]

a) heavy water b) cold water
c) cadmium d) carbon

15. the material used for moderator is [d]

a) heavy water b) cold water
c) cadmium d) graphite

16. A surge tank provided near [d]

a) dam b) trash rack
c) spillway d) turbine

17.the function of surge tank [c]

a) supply water with pressure b) produce surge in pipe
c) relieve water hammer effect d) none

18. water hammer developed in [c]

a) surge tank b) trash rack
c) penstock d) turbine

19. in hydral plant, which one is used to carrying water from intake to turbine [c]

a) surge tank b) trash rack
c) penstock d) turbine

20. which fuels used in nuclear plant [c]

a) uranium b) thorium
c) both d) none

21. the predominant source of energy on earth is [b]

a) moon b) sun
c) plants d) none

22. which material is mostly used in solar cell [d]

a) silver b) iron
c) aluminium d) silicon

23. the output of solar cell depends on [a]

a) intensity of solar radiation b) ultraviolet radiation
c) infrared radiation d) none

24.solar photovoltaic cell converts solar energy into….energy [c]

a) mechanical b) heat
c) electrical d) none

25.solar energy travels through space by process of [d]

a) conduction b) convection
c) transportation d) radiation

26. sunlight reaches the earth through….. radiation [d]

a) direct b) diffuse
c) scattered d) all

27.what kind of energy does a wind turbine use [a]

a) kinetic energy b) potential energy
c) chemical energy d) thermal energy

28. wind blows because of a difference in [a]

a) temperature b) lattitude
c) longitude d) height

29.with increase in height, wind speed [a]

a) increases b) decreases
c) constant d) none

30.anemometer used to measure [b]

a) solar radiation b) wind speed
c) temperature d) ocean depth

31.which component used to carry the power from one place to another place [c]

a) cross arms b) insulator
c) conductor d) supporter

32. supports also called [c]

a) towers b) poles
c) a or b d) none

33. commonly using conductors are [d]

a) copper b) aluminum
c) galvanized steel d) all

34. cross arms are used to support the [b]

a) dampers b) insulator
c) conductor d) supports

35. which one is used to protect the transmission line from lightning's [d]

a) cross arms b) dampers
c) conductor d) arresters

36. which are attached to supports and insulate the conductors from the ground[b]

a) cross arms b) insulator
c) conductor d) supporter

37. secondary distribution system employs [b]

a) 3phase 3 wire b) 3phase 4 wire
c) both d) none

38. primary distribution system employs [a]

a) 3phase 3 wire b) 3phase 4 wire
c) both d) none

39. the usual generating voltage is [c]

a) 132kv b) 33kv
c) 11kv d) 6.6kv

40. which plant is not using turbine and alternator [d]

a) thermal plant b) wind plant
c) hydral plant d) solar plant

10. Assignments

S.No	Question	BL	CO
1	Draw a typical layout of a thermal power plant and describe the function of each component.	1	1
2	Draw a neat schematic diagram of a Hydro-electric plant and write the functions of various components.	2	4
3	Explain about Solar & Wind Generation Systems?	2	3
4	Explain about primary & secondary of Distribution Systems	2	4

11. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	**Define Penstock.** It is a huge steel pipe which carries water from reservoir to turbine. The Potential energy is converted into Mechanical energy.	1	1
2	**Define Surge Tank.** It connected between reservoir and Turbine. It maintains equal pressure difference in pipes without burst.	1	1
3	**What is the purpose of Boiler in Thermal Power Plant?** The coal is taken into boiler and burnt to produce heat energy. This heat energy is utilized to convert water into high pressure and High temperature steam.	1	4
4	**What is the purpose of Condenser in Thermal Power Plant?** The exhausted steam from turbine is converted into water by cold water circulation.	1	4
5	**What are the different types of Distribution Systems.** 1. Primary conversion 33KV to 11 KV **2.** Secondary conversion 11KV to 0.4KV	1	4
6	**What is meant by Complex power?** Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power, P and its imaginary part is reactive power, Q. and S = P + jQ	1	4
7	**Define Phasor and Phase angle.** A sinusoidal wave form canbe represented or in terms of a Phasor. A Phasor is a vector with definite magnitude and direction. From the Phasor the sinusoidal wave form can be reconstructed. Phase angle is the angular measurement that specifies the position of the alternating quantity relative to a reference	2	3

8	**Write the Algorithm for Nodal Analysis.** Select a node as the reference node. Assign voltages V1,V2,…Vn-1 to the remaining n-1 nodes. Apply KCL to each of the n-1 nodes. Solve the resulting simultaneous equations to obtain the unknown node voltages.	1	3
9	**Distinguish between a Loop & Mesh of a circuit (DEC, '10)** The closed path of a network is called a Loop. An elementary form of a loop which cannot be further divided is called a mesh. In other words Mesh is closed path does not contain an other loop within it.	1	3
10	**Why transmission voltage is at higher levels?** In order to reduce transmission losses power is transmitted at higher voltages of 132KV or 220KV	1	2

12. Part B- Questions

S.No	Question	BL	CO
1	Draw a typical layout of a thermal power plant and describe the function of each component.	1	1
2	Draw a neat schematic diagram of a Hydro-electric plant and write the functions of various components.	2	3
3	Explain about Solar & Wind Generation Systems?	3	4
4	With a neat schematic diagram, explain the operation of a Nuclear power plant. Also discuss the advantages and disadvantages of it.	2	4
5	Explain about primary & secondary of Distribution Systems	3	4

13. Supportive Online Certification Courses

1. Basic Electrical Circuits by Prof. Ankush Sharma, conducted by IIT Kanpur – 4 weeks
2. Basic Electrical Engineering By Prof. Bhattacharya, conducted by IIT Madras – 08 weeks.

14. Real Time Applications

S.No	Application	CO
1	In electrical engineering, single-phase electric power is the distribution of alternating current electric power using a system in which all the voltages of the supply vary in unison. Single-phase distribution is used when loads are mostly lighting and heating, with few large electric motors.	4
2	Single phase supply is used for small electrical appliances and equipment, mostly for home applications.	4
3	All 230 V 50 Hz supply is used in electrical appliances like TV, EV charger level 2, fan, tubelight, LED bulb etc.	4
4	In India single phase is used for converting electrical energy into other useful forms like rotation (motion), light, heat , sound etc.Usually this is 3 to 5 KW.	4

15. Contents Beyond the Syllabus

1. Electricity generation from Geothermal Hotspots
2. Economics of power supply system

16. Prescribed Text Books & Reference Books

Text Books:

1. D. P. Kothari and I. J. Nagrath - "Basic Electrical Engineering" - Tata McGraw Hill -2010.
2. V.K. Mehta & Rohit Mehta, "Principles of Power System" – S.Chand – 2018.

References:

1. L. S. Bobrow - "Fundamentals of Electrical Engineering" - Oxford University Press -2011.
2. E. Hughes - "Electrical and Electronics Technology" - Pearson - 2010.

3. C.L. Wadhwa – “Generation Distribution and Utilization of Electrical Energy”, 3rd Edition, New Age International Publications.

17. Mini Project Suggestion

1. **Temperature Controlled System:**
This project shows a temperature-controlled system. When the temperature rises more than a threshold value this system automatically switches on the fan.
2. **Conversion of Single Phase to Three Phase Supply:**
This paper shows a converter that converts the single-phase supply into a three-phase supply using thyristors. Go through the paper for more information.

3. **Smoke Detector Alarm Circuit:**
Here is the circuit showing a smoke detector alarm. This can also be used to indicate the fire. This circuit uses a smoke detector and an LM358 Comparator.

4. **Railway Security System based on Wireless Sensor Networks:**
This paper describes different methods for detecting the defects in railway tracks and methods for maintaining the track are also proposed.

5. **Control Electrical Devices From your android phone:**
This paper shows the controlling of electrical devices from an android phone using an app. This project uses Arduino for controlling the devices.